Chapter 6

Best Linear Unbiased Estimators

6.1 Introduction

It frequently occurs in practice that the MVU estimator, if it exists, cannot be found. For instance, we may not know the PDF of the data or even be willing to assume a model for it. In this case our previous methods, which rely on the CRLB and the theory of sufficient statistics, cannot be applied. Even if the PDF is known, the latter approaches are not guaranteed to produce the MVU estimator. Faced with our inability to determine the optimal MVU estimator, it is reasonable to resort to a suboptimal estimator. In doing so, we are never sure how much performance we may have lost (since the minimum variance of the MVU estimator is unknown). However, if the variance of the suboptimal estimator can be ascertained and if it meets our system specifications, then its use may be justified as being adequate for the problem at hand. If its variance is too large, then we will need to look at other suboptimal estimators, hoping to find one that meets our specifications. A common approach is to restrict the estimator to be *linear* in the data and find the linear estimator that is *unbiased* and has *minimum variance*. As we will see, this estimator, which is termed the *best linear unbiased estimator* (BLUE), can be determined with knowledge of only the first and second moments of the PDF. Since complete knowledge of the PDF is not necessary, the BLUE is frequently more suitable for practical implementation.

6.2 Summary

The BLUE is based on the linear estimator defined in (6.1). If we constrain this estimator to be unbiased as in (6.2) and to minimize the variance of (6.3), then the BLUE is given by (6.5). The minimum variance of the BLUE is (6.6). To determine the BLUE requires knowledge of only the mean and covariance of the data. In the vector parameter case the BLUE is given by (6.16) and the covariance by (6.17). In either the scalar parameter or vector parameter case, if the data are Gaussian, then the BLUE is also the MVU estimator.

6.3 Definition of the BLUE

We observe the data set $\{x[0], x[1], \ldots, x[N-1]\}$ whose PDF $p(\mathbf{x};\theta)$ depends on an unknown parameter θ. The BLUE restricts the estimator to be linear in the data or

$$\hat{\theta} = \sum_{n=0}^{N-1} a_n x[n] \tag{6.1}$$

where the a_n's are constants yet to be determined. (See also Problem 6.6 for a similar definition except for the addition of a constant.) Depending on the a_n's chosen, we may generate a large number of different estimators of θ. However, the best estimator or BLUE is defined to be the one that is unbiased and has minimum variance. Before determining the a_n's that yield the BLUE, some comments about the optimality of the BLUE are in order. Since we are restricting the class of estimators to be linear, the BLUE will be optimal (that is to say, the MVU estimator) only when the MVU estimator turns out to be linear. For example, for the problem of estimating the value of a DC level in WGN (see Example 3.3) the MVU estimator is the sample mean

$$\hat{\theta} = \bar{x} = \sum_{n=0}^{N-1} \frac{1}{N} x[n]$$

which is clearly linear in the data. Hence, if we restrict our attention to only linear estimators, then we will lose nothing since the MVU estimator is within this class. Figure 6.1a depicts this idea. On the other hand, for the problem of estimating the mean of uniformly distributed noise (see Example 5.8), the MVU estimator was found to be

$$\hat{\theta} = \frac{N+1}{2N} \max\, x[n]$$

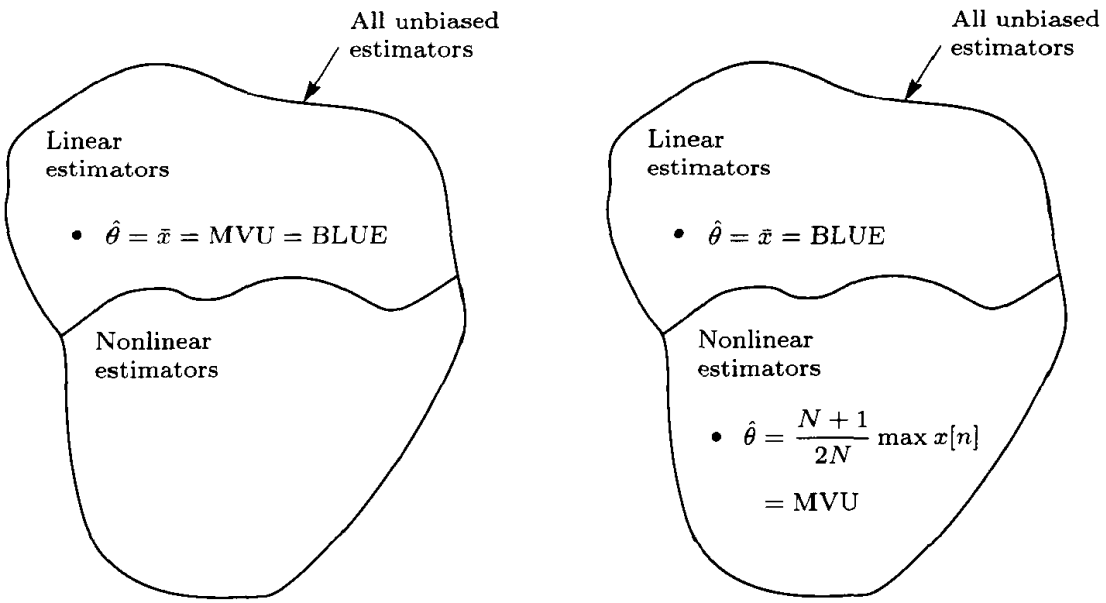
which is nonlinear in the data. If we restrict our estimator to be linear, then the BLUE is the sample mean, as we will see shortly. The BLUE for this problem is suboptimal, as illustrated in Figure 6.1b. As further shown in Example 5.8, the difference in performance is substantial. Unfortunately, without knowledge of the PDF there is no way to determine the loss in performance by resorting to a BLUE.

Finally, for some estimation problems the use of a BLUE can be totally inappropriate. Consider the estimation of the power of WGN. It is easily shown that the MVU estimator is (see Example 3.6)

$$\hat{\sigma^2} = \frac{1}{N} \sum_{n=0}^{N-1} x^2[n]$$

which is nonlinear in the data. If we force the estimator to be linear as per (6.1), so that

$$\hat{\sigma^2} = \sum_{n=0}^{N-1} a_n x[n],$$



(a) DC level in WGN; BLUE is optimal

(b) Mean of uniform noise; BLUE is suboptimal

Figure 6.1 Optimality of BLUE

the expected value of the estimator becomes

$$E(\hat{\sigma^2}) = \sum_{n=0}^{N-1} a_n E(x[n]) = 0$$

since $E(x[n]) = 0$ for all n. Thus, we cannot even find a single linear estimator that is unbiased, let alone one that has minimum variance. Although the BLUE is unsuitable for this problem, a BLUE utilizing the *transformed data* $y[n] = x^2[n]$ would produce a viable estimator since for

$$\hat{\sigma^2} = \sum_{n=0}^{N-1} a_n y[n] = \sum_{n=0}^{N-1} a_n x^2[n]$$

the unbiased constraint yields

$$E(\hat{\sigma^2}) = \sum_{n=0}^{N-1} a_n \sigma^2 = \sigma^2.$$

There are many values of the a_n's that could satisfy this constraint. Can you guess what the a_n's should be to yield the BLUE? (See also Problem 6.5 for an example of this data transformation approach.) Hence, with enough ingenuity the BLUE may still be used if the data are first transformed suitably.

6.4 Finding the BLUE

To determine the BLUE we constrain $\hat{\theta}$ to be linear and unbiased and then find the a_n's to minimize the variance. The unbiased constraint, is from (6.1),

$$E(\hat{\theta}) = \sum_{n=0}^{N-1} a_n E(x[n]) = \theta. \tag{6.2}$$

The variance of $\hat{\theta}$ is

$$\text{var}(\hat{\theta}) = E\left[\left(\sum_{n=0}^{N-1} a_n x[n] - E\left(\sum_{n=0}^{N-1} a_n x[n]\right)\right)^2\right].$$

But by using (6.2) and letting $\mathbf{a} = [a_0\, a_1 \ldots a_{N-1}]^T$ we have

$$\begin{aligned} \text{var}(\hat{\theta}) &= E\left[\left(\mathbf{a}^T\mathbf{x} - \mathbf{a}^T E(\mathbf{x})\right)^2\right] \\ &= E\left[\left(\mathbf{a}^T(\mathbf{x} - E(\mathbf{x}))\right)^2\right] \\ &= E\left[\mathbf{a}^T(\mathbf{x} - E(\mathbf{x}))(\mathbf{x} - E(\mathbf{x}))^T\mathbf{a}\right] \\ &= \mathbf{a}^T\mathbf{C}\mathbf{a}. \end{aligned} \tag{6.3}$$

The vector $\mathbf{a}$ of weights is found by minimizing (6.3) subject to the constraint of (6.2). Before proceeding we need to assume some form for $E(x[n])$. In order to satisfy the unbiased constraint, $E(x[n])$ must be linear in θ or

$$E(x[n]) = s[n]\theta \tag{6.4}$$

where the $s[n]$'s are *known*. Otherwise, it may be impossible to satisfy the constraint. As an example, if $E(x[n]) = \cos\theta$, then the unbiased constraint is $\sum_{n=0}^{N-1} a_n \cos\theta = \theta$ for all θ. Clearly, there do not exist a_n's that will satisfy the unbiased constraint. Note that if we write $x[n]$ as

$$x[n] = E(x[n]) + [x[n] - E(x[n])]$$

then by viewing $x[n] - E(x[n])$ as noise or $w[n]$ we have

$$x[n] = \theta s[n] + w[n].$$

The assumption of (6.4) means that *the BLUE is applicable to amplitude estimation of known signals in noise.* To generalize its use we require a nonlinear transformation of the data as already described.

With the assumption given by (6.4) we now summarize the estimation problem. To find the BLUE we need to minimize the variance

$$\text{var}(\hat{\theta}) = \mathbf{a}^T\mathbf{C}\mathbf{a}$$

subject to the unbiased constraint, which from (6.2) and (6.4) becomes

$$\begin{aligned}\sum_{n=0}^{N-1} a_n E(x[n]) &= \theta \\ \sum_{n=0}^{N-1} a_n s[n]\theta &= \theta \\ \sum_{n=0}^{N-1} a_n s[n] &= 1\end{aligned}$$

or

$$\mathbf{a}^T\mathbf{s} = 1$$

where $\mathbf{s} = [s[0]\, s[1] \ldots s[N-1]]^T$. The solution to this minimization problem is derived in Appendix 6A as

$$\mathbf{a}_{\text{opt}} = \frac{\mathbf{C}^{-1}\mathbf{s}}{\mathbf{s}^T\mathbf{C}^{-1}\mathbf{s}}$$

so that the BLUE is

$$\hat{\theta} = \frac{\mathbf{s}^T\mathbf{C}^{-1}\mathbf{x}}{\mathbf{s}^T\mathbf{C}^{-1}\mathbf{s}} \tag{6.5}$$

and has minimum variance

$$\text{var}(\hat{\theta}) = \frac{1}{\mathbf{s}^T\mathbf{C}^{-1}\mathbf{s}}. \tag{6.6}$$

Note from (6.4) that since $E(\mathbf{x}) = \theta\mathbf{s}$, the BLUE is unbiased

$$\begin{aligned}E(\hat{\theta}) &= \frac{\mathbf{s}^T\mathbf{C}^{-1}E(\mathbf{x})}{\mathbf{s}^T\mathbf{C}^{-1}\mathbf{s}} \\ &= \frac{\mathbf{s}^T\mathbf{C}^{-1}\theta\mathbf{s}}{\mathbf{s}^T\mathbf{C}^{-1}\mathbf{s}} \\ &= \theta.\end{aligned}$$

Also, as we asserted earlier, to determine the BLUE we only require knowledge only of

1. s or the scaled mean

2. C, the covariance

or the first two moments but not the entire PDF. Some examples now follow.

Example 6.1 - DC Level in White Noise

If we observe

$$x[n] = A + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is white noise with variance σ^2 (*and of unspecified PDF*), then the problem is to estimate A. Since $w[n]$ is not necessarily Gaussian, the noise samples may be

statistically dependent even though they are uncorrelated. Because $E(x[n]) = A$, we have from (6.4) that $s[n] = 1$ and therefore $\mathbf{s} = \mathbf{1}$. From (6.5) the BLUE is

$$\begin{aligned} \hat{A} &= \frac{\mathbf{1}^T \frac{1}{\sigma^2}\mathbf{I}\mathbf{x}}{\mathbf{1}^T \frac{1}{\sigma^2}\mathbf{I}\mathbf{1}} \\ &= \frac{1}{N}\sum_{n=0}^{N-1} x[n] = \bar{x} \end{aligned}$$

and has minimum variance, from (6.6),

$$\begin{aligned} \text{var}(\hat{A}) &= \frac{1}{\mathbf{1}^T \frac{1}{\sigma^2}\mathbf{1}} \\ &= \frac{\sigma^2}{N}. \end{aligned}$$

Hence, *the sample mean is the BLUE independent of the PDF of the data.* In addition, as already discussed, it is the MVU estimator for Gaussian noise. ◇

Example 6.2 - DC Level in Uncorrelated Noise

Now let $w[n]$ be zero mean *uncorrelated* noise with $\text{var}(w[n]) = \sigma_n^2$ and repeat Example 6.1. As before, $\mathbf{s} = \mathbf{1}$, and from (6.5) and (6.6)

$$\begin{aligned} \hat{A} &= \frac{\mathbf{1}^T\mathbf{C}^{-1}\mathbf{x}}{\mathbf{1}^T\mathbf{C}^{-1}\mathbf{1}} \\ \text{var}(\hat{A}) &= \frac{1}{\mathbf{1}^T\mathbf{C}^{-1}\mathbf{1}}. \end{aligned}$$

The covariance matrix is

$$\mathbf{C} = \begin{bmatrix} \sigma_0^2 & 0 & \dots & 0 \\ 0 & \sigma_1^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_{N-1}^2 \end{bmatrix},$$

and its inverse is

$$\mathbf{C}^{-1} = \begin{bmatrix} \frac{1}{\sigma_0^2} & 0 & \dots & 0 \\ 0 & \frac{1}{\sigma_1^2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \frac{1}{\sigma_{N-1}^2} \end{bmatrix}.$$

Therefore,

$$\hat{A} = \frac{\sum_{n=0}^{N-1} \frac{x[n]}{\sigma_n^2}}{\sum_{n=0}^{N-1} \frac{1}{\sigma_n^2}} \tag{6.7}$$

$$\text{var}(\hat{A}) = \frac{1}{\sum_{n=0}^{N-1} \frac{1}{\sigma_n^2}}. \tag{6.8}$$

The BLUE weights those samples most heavily with smallest variances in an attempt to equalize the noise contribution from each sample. The denominator in (6.7) is the scale factor needed to make the estimator unbiased. See also Problem 6.2 for some further results. ◇

In general, the presence of $\mathbf{C}^{-1}$ in the BLUE acts to prewhiten the data prior to averaging. This was previously encountered in Example 4.4 which discussed estimation of a DC level in colored *Gaussian* noise. Also, in that example the identical estimator for A was obtained. Because of the Gaussian noise assumption, however, the estimator could be said to be efficient and hence MVU. It is not just coincidental that the MVU estimator for Gaussian noise and the BLUE are identical for this problem. This is a general result which says that for estimation of the parameters of a *linear model* (see Chapter 4) the BLUE is identical to the MVU estimator for Gaussian noise. We will say more about this in the next section.

6.5 Extension to a Vector Parameter

If the parameter to be estimated is a $p \times 1$ vector parameter, then for the estimator to be linear in the data we require

$$\hat{\theta}_i = \sum_{n=0}^{N-1} a_{in} x[n] \qquad i = 1, 2, \ldots, p \tag{6.9}$$

where the a_{in}'s are weighting coefficients. In matrix form this is

$$\hat{\boldsymbol{\theta}} = \mathbf{A}\mathbf{x}$$

where $\mathbf{A}$ is a $p \times N$ matrix. In order for $\hat{\boldsymbol{\theta}}$ to be unbiased we require

$$E(\hat{\theta}_i) = \sum_{n=0}^{N-1} a_{in} E(x[n]) = \theta_i \qquad i = 1, 2, \ldots, p \tag{6.10}$$

or in matrix form

$$E(\hat{\boldsymbol{\theta}}) = \mathbf{A}E(\mathbf{x}) = \boldsymbol{\theta}. \tag{6.11}$$

Keep in mind that a linear estimator will be appropriate only for problems in which the unbiased constraint can be satisfied. From (6.11) we must have that

$$E(\mathbf{x}) = \mathbf{H}\boldsymbol{\theta} \tag{6.12}$$

for $\mathbf{H}$ a known $N \times p$ matrix. In the scalar parameter case (see (6.4))

$$E(\mathbf{x}) = \underbrace{\begin{bmatrix} s[0] \\ s[1] \\ \vdots \\ s[N-1] \end{bmatrix}}_{\mathbf{H}} \theta$$

so that (6.12) is truly the vector parameter generalization. Now, substitution of (6.12) into (6.11) produces the unbiased constraint

$$\mathbf{AH} = \mathbf{I}. \tag{6.13}$$

If we define $\mathbf{a}_i = [a_{i0}\, a_{i1} \ldots a_{i(N-1)}]^T$, so that $\hat{\theta}_i = \mathbf{a}_i^T\mathbf{x}$, the unbiased constraint may be rewritten for each $\mathbf{a}_i$ by noting that

$$\mathbf{A} = \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \vdots \\ \mathbf{a}_p^T \end{bmatrix}$$

and letting $\mathbf{h}_i$ denote the ith column of $\mathbf{H}$, so that

$$\mathbf{H} = \begin{bmatrix} \mathbf{h}_1 & \mathbf{h}_2 & \ldots & \mathbf{h}_p \end{bmatrix}.$$

With these definitions the unbiased constraint of (6.13) reduces to

$$\mathbf{a}_i^T\mathbf{h}_j = \delta_{ij} \qquad i = 1, 2, \ldots, p; j = 1, 2, \ldots, p. \tag{6.14}$$

Also, the variance is

$$\text{var}(\hat{\theta}_i) = \mathbf{a}_i^T\mathbf{C}\mathbf{a}_i \tag{6.15}$$

using the results for the scalar case (see (6.3)). The BLUE for a vector parameter is found by minimizing (6.15) subject to the constraints of (6.14), repeating the minimization for each i. In Appendix 6B this minimization is carried out to yield the BLUE as

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{C}^{-1}\mathbf{x} \tag{6.16}$$

and a covariance matrix of

$$\mathbf{C}_{\hat{\theta}} = (\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}. \tag{6.17}$$

The form of the BLUE is identical to the MVU estimator for the general linear model (see (4.25)). This extends the observations of Examples 6.1 and 6.2. That this should be the case follows from the definition of the general linear model

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w}$$

where $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{C})$. With this model we are attempting to estimate $\boldsymbol{\theta}$, where $E(\mathbf{x}) = \mathbf{H}\boldsymbol{\theta}$. This is exactly the assumption made in (6.12). But from (4.25) the MVU estimator for Gaussian data is

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{C}^{-1}\mathbf{x}$$

which is clearly linear in $\mathbf{x}$. Hence, restricting our estimator to be linear does not lead to a suboptimal estimator since the MVU estimator is within the linear class. We may conclude that *if the data are truly Gaussian, then the BLUE is also the MVU estimator.*

To summarize our discussions we now state the general BLUE for a vector parameter of a general linear model. In the present context the general linear model does not assume Gaussian noise. We refer to the data as having the general linear model *form.* The following theorem is termed the *Gauss-Markov theorem.*

Theorem 6.1 (Gauss-Markov Theorem) *If the data are of the general linear model form*

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w} \tag{6.18}$$

where $\mathbf{H}$ *is a known* $N \times p$ *matrix,* $\boldsymbol{\theta}$ *is a* $p \times 1$ *vector of parameters to be estimated, and* $\mathbf{w}$ *is a* $N \times 1$ *noise vector with zero mean and covariance* $\mathbf{C}$ *(the PDF of* $\mathbf{w}$ *is otherwise arbitrary), then the BLUE of* $\boldsymbol{\theta}$ *is*

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{C}^{-1}\mathbf{x} \tag{6.19}$$

and the minimum variance of $\hat{\theta}_i$ *is*

$$\text{var}(\hat{\theta}_i) = \left[(\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}\right]_{ii}. \tag{6.20}$$

In addition, the covariance matrix of $\hat{\boldsymbol{\theta}}$ *is*

$$\mathbf{C}_{\hat{\theta}} = (\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}. \tag{6.21}$$

Some properties of BLUEs are given in Problems 6.12 and 6.13. Further examples of the computation of BLUEs are given in Chapter 4.

6.6 Signal Processing Example

In the design of many signal processing systems we are given measurements that may not correspond to the "raw data." For instance, in optical interferometry the input to the system is light, and the output is the photon count as measured by a photodetector [Chamberlain 1979]. The system is designed to mix the incoming light signal with a delayed version of itself (through spatially separated mirrors and a nonlinear device).

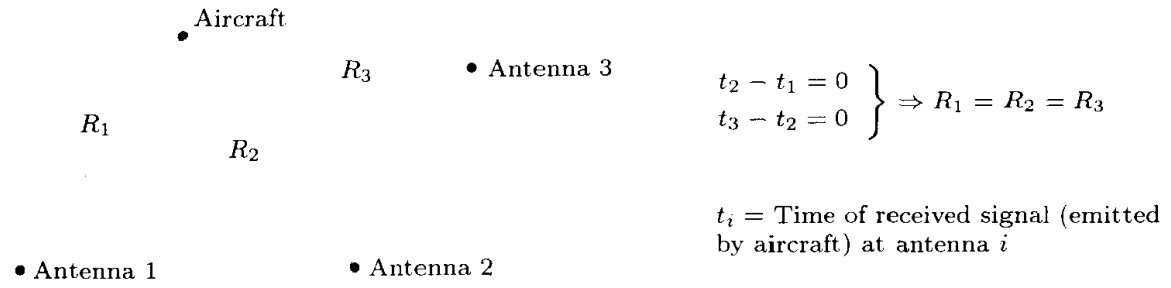


Figure 6.2 Localization of aircraft by time difference of arrival measurements

The data then is more nearly modeled as an autocorrelation function. The raw data, that is, the light intensity, is unavailable. From the given information it is desired to estimate the spectrum of the incoming light. To assume that the measurements are Gaussian would probably be unrealistic. This is because if a random process is Gaussian, then the autocorrelation data are certainly non-Gaussian [Anderson 1971]. Furthermore, the exact PDF is impossible to find in general, being mathematically intractable. Such a situation lends itself to the use of the BLUE. A second example is explored in some detail next.

Example 6.3 - Source Localization

A problem of considerable interest (especially to anyone who has traveled by airplane) is that of determining the position of a source based on the emitted signal of the source. In order to track the position of an aircraft we typically use several antennas. A common method as shown in Figure 6.2 is to base our estimate on the time difference of arrival (TDOA) measurements. In the figure the signal emitted by the aircraft is received at the same time at all three antennas, so that $t_1 = t_2 = t_3$. The TDOA $t_2 - t_1$ between antennas 1 and 2 is zero, as is the TDOA $t_3 - t_2$ between antennas 2 and 3. Hence, we may conclude that the ranges R_i to all three antennas are the same. The dashed line between antennas 1 and 2 indicates the possible positions of the aircraft in order for $R_1 = R_2$, and likewise for the dashed line between antennas 2 and 3. The intersection of these lines gives the aircraft position. In the more general situation for which the TDOAs are not zero, the dashed lines become hyperbolas, the intersection again giving the position. It is necessary to have at least three antennas for localization, and when noise is present, it is desirable to have more.

We now examine the localization problem using estimation theory [Lee 1975]. To do so we assume that N antennas have been placed at known locations and that the time of arrival measurements t_i for $i = 0, 1, \ldots, N-1$ are available. The problem is to estimate the source position (x_s, y_s) as illustrated in Figure 6.3. The arrival times are assumed to be corrupted by noise with zero mean and known covariance but otherwise

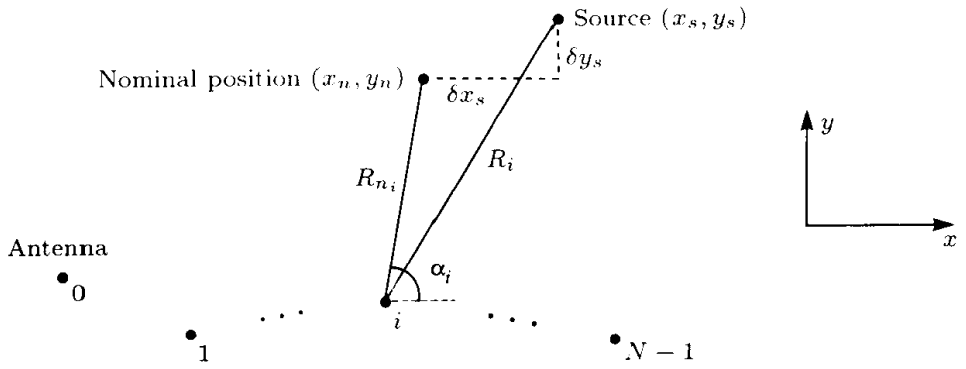


Figure 6.3 Source localization geometry

unknown PDF. For a signal emitted by the source at time $t = T_0$, the measurements are modeled by

$$t_i = T_0 + R_i/c + \epsilon_i \qquad i = 0, 1, \ldots, N-1 \tag{6.22}$$

where the ϵ_i's are measurement noises and c denotes the propagation speed. The noise samples are assumed to be zero mean with variance σ^2 and uncorrelated with each other. *No assumptions are made about the PDF of the noise.* To proceed further we must relate the range from each antenna R_i to the unknown position $\boldsymbol{\theta} = [x_s\, y_s]^T$. Letting the position of the ith antenna be (x_i, y_i), which is assumed to be known, we have that

$$R_i = \sqrt{(x_s - x_i)^2 + (y_s - y_i)^2}. \tag{6.23}$$

Substituting (6.23) into (6.22), we see that the model is nonlinear in the unknown parameters x_s and y_s. To apply the Gauss-Markov theorem we will assume that a nominal source position (x_n, y_n) is available. This nominal position, which is close to the true source position, could have been obtained from previous measurements. Such a situation is typical when the source is being tracked. Hence, to estimate the *new* source position we require an estimate of $\boldsymbol{\theta} = [(x_s - x_n)\ (y_s - y_n)]^T = [\delta x_s\, \delta y_s]^T$, as shown in Figure 6.3. Denote the nominal ranges by R_{n_i}. We use a first-order Taylor expansion of R_i (considered as a two-dimensional function of x_s and y_s) about the nominal position $x_s = x_n, y_s = y_n$

$$R_i \approx R_{n_i} + \frac{x_n - x_i}{R_{n_i}}\delta x_s + \frac{y_n - y_i}{R_{n_i}}\delta y_s. \tag{6.24}$$

See also Chapter 8 for a more complete description of the linearization of a nonlinear model. With this approximation we have upon substitution in (6.22)

$$t_i = T_0 + \frac{R_{n_i}}{c} + \frac{x_n - x_i}{R_{n_i}c}\delta x_s + \frac{y_n - y_i}{R_{n_i}c}\delta y_s + \epsilon_i$$

which is now linear in the unknown parameters δx_s and δy_s. Alternatively, since as defined in Figure 6.3

$$\frac{x_n - x_i}{R_{n_i}} = \cos\alpha_i$$

$$\frac{y_n - y_i}{R_{n_i}} = \sin\alpha_i,$$

the model simplifies to

$$t_i = T_0 + \frac{R_{n_i}}{c} + \frac{\cos\alpha_i}{c}\delta x_s + \frac{\sin\alpha_i}{c}\delta y_s + \epsilon_i.$$

The term R_{n_i}/c is a *known* constant which can be incorporated into the measurement by letting (see also Problem 6.6)

$$\tau_i = t_i - \frac{R_{n_i}}{c}.$$

Hence, we have for our linear model

$$\tau_i = T_0 + \frac{\cos\alpha_i}{c}\delta x_s + \frac{\sin\alpha_i}{c}\delta y_s + \epsilon_i \tag{6.25}$$

where the unknown parameters are $T_0, \delta x_s, \delta y_s$. By assuming the time T_0 when the source emits a signal is unknown, we are bowing to practical considerations. Knowledge of T_0 would require accurate clock synchronization between the source and the receiver, which for economic reasons is avoided. It is customary to consider time *difference* of arrivals or TDOA measurements to eliminate T_0 in (6.25). We generate the TDOAs as

$$\begin{aligned} \xi_1 &= \tau_1 - \tau_0 \\ \xi_2 &= \tau_2 - \tau_1 \\ &\vdots \\ \xi_{N-1} &= \tau_{N-1} - \tau_{N-2}. \end{aligned}$$

Then, from (6.25) we have as our final linear model

$$\xi_i = \frac{1}{c}(\cos\alpha_i - \cos\alpha_{i-1})\delta x_s + \frac{1}{c}(\sin\alpha_i - \sin\alpha_{i-1})\delta y_s + \epsilon_i - \epsilon_{i-1} \tag{6.26}$$

for $i = 1, 2, \ldots, N-1$. An alternative but equivalent approach is to use (6.25) and estimate T_0 as well as the source position. Now we have reduced the estimation problem to that described by the Gauss-Markov theorem where

$$\begin{aligned} \boldsymbol{\theta} &= \begin{bmatrix} \delta x_s & \delta y_s \end{bmatrix}^T \\ \mathbf{H} &= \frac{1}{c}\begin{bmatrix} \cos\alpha_1 - \cos\alpha_0 & \sin\alpha_1 - \sin\alpha_0 \\ \cos\alpha_2 - \cos\alpha_1 & \sin\alpha_2 - \sin\alpha_1 \\ \vdots & \vdots \\ \cos\alpha_{N-1} - \cos\alpha_{N-2} & \sin\alpha_{N-1} - \sin\alpha_{N-2} \end{bmatrix} \\ \mathbf{w} &= \begin{bmatrix} \epsilon_1 - \epsilon_0 \\ \epsilon_2 - \epsilon_1 \\ \vdots \\ \epsilon_{N-1} - \epsilon_{N-2} \end{bmatrix}. \end{aligned}$$

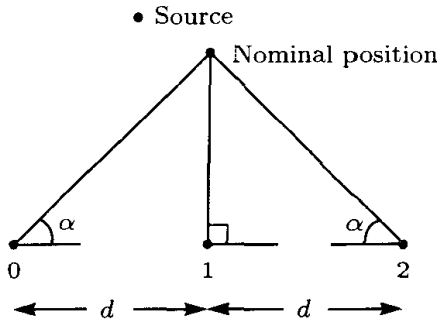


Figure 6.4 Example of source localization geometry for line array with minimum number of antennas

The noise vector $\mathbf{w}$ is zero mean but is no longer composed of uncorrelated random variables. To find the covariance matrix note that

$$\mathbf{w} = \underbrace{\begin{bmatrix} -1 & 1 & 0 & 0 & \dots & 0 \\ 0 & -1 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & 0 & -1 & 1 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} \epsilon_0 \\ \epsilon_1 \\ \vdots \\ \epsilon_{N-1} \end{bmatrix}}_{\boldsymbol{\epsilon}},$$

where $\mathbf{A}$ has dimension $(N-1) \times N$. Since the covariance matrix of $\boldsymbol{\epsilon}$ is $\sigma^2\mathbf{I}$, we have that

$$\mathbf{C} = E[\mathbf{A}\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T\mathbf{A}^T] = \sigma^2\mathbf{A}\mathbf{A}^T.$$

From (6.19) the BLUE of the source position parameters is

$$\begin{aligned} \hat{\boldsymbol{\theta}} &= (\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{C}^{-1}\boldsymbol{\xi} \\ &= [\mathbf{H}^T(\mathbf{A}\mathbf{A}^T)^{-1}\mathbf{H}]^{-1}\mathbf{H}^T(\mathbf{A}\mathbf{A}^T)^{-1}\boldsymbol{\xi} \end{aligned} \tag{6.27}$$

and the minimum variance, is from (6.20),

$$\text{var}(\hat{\theta}_i) = \sigma^2 \left[\{\mathbf{H}^T(\mathbf{A}\mathbf{A}^T)^{-1}\mathbf{H}\}^{-1}\right]_{ii} \tag{6.28}$$

or the covariance matrix, is from (6.21),

$$\mathbf{C}_{\hat{\theta}} = \sigma^2 \left[\mathbf{H}^T(\mathbf{A}\mathbf{A}^T)^{-1}\mathbf{H}\right]^{-1}. \tag{6.29}$$

As an example, for the three antenna line array (the minimum number of antennas required) shown in Figure 6.4 we have that

$$\begin{aligned} \mathbf{H} &= \frac{1}{c}\begin{bmatrix} -\cos\alpha & 1-\sin\alpha \\ -\cos\alpha & -(1-\sin\alpha) \end{bmatrix} \\ \mathbf{A} &= \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}. \end{aligned}$$

After substituting in (6.29) we have for the covariance matrix

$$\mathbf{C}_{\hat{\theta}} = \sigma^2 c^2 \begin{bmatrix} \frac{1}{2\cos^2\alpha} & 0 \\ 0 & \frac{3/2}{(1-\sin\alpha)^2} \end{bmatrix}.$$

For the best localization we would like α to be small. This is accomplished by making the spacing between antennas d large, so that the *baseline* of the array, which is the total length, is large. Note also that the localization accuracy is range dependent, with the best accuracy for short ranges or for α small.

References

Anderson, T.W., *The Statistical Analysis of Time Series*, J. Wiley, New York, 1971.

Chamberlain, J., *The Principles of Interferometric Spectroscopy*, J. Wiley, New York, 1979.

Lee, H.B., "A Novel Procedure for Assessing the Accuracy of Hyperbolic Multilateration Systems," *IEEE Trans. Aerosp. Electron. Syst.*, Vol. 11, pp. 2–15, Jan. 1975.

Problems

6.1 If $x[n] = Ar^n + w[n]$ for $n = 0, 1, \ldots, N-1$, where A is an unknown parameter, r is a known constant, and $w[n]$ is zero mean white noise with variance σ^2, find the BLUE of A and the minimum variance. Does the minimum variance approach zero as $N \to \infty$?

6.2 In Example 6.2 let the noise variances be given by $\sigma_n^2 = n + 1$ and examine what happens to the variance of the BLUE as $N \to \infty$. Repeat for $\sigma_n^2 = (n+1)^2$ and explain your results.

6.3 Consider the estimation problem described in Example 6.2 but now assume that the noise samples are correlated with the covariance matrix

$$\mathbf{C} = \sigma^2 \begin{bmatrix} \begin{matrix} 1 & \rho \\ \rho & 1 \end{matrix} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \begin{matrix} 1 & \rho \\ \rho & 1 \end{matrix} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \begin{matrix} 1 & \rho \\ \rho & 1 \end{matrix} \end{bmatrix}$$

where $|\rho| < 1$ and N, the dimension of the matrix, is assumed to be even. $\mathbf{C}$ is a block-diagonal matrix and so is easily inverted (see (Appendix 1)). Find the BLUE and its variance and interpret your results.

6.4 The observed samples $\{x[0], x[1], \ldots, x[N-1]\}$ are IID according to the following PDFs:

a. Laplacian

$$p(x[n]; \mu) = \frac{1}{2} \exp[-|x[n] - \mu|]$$

b. Gaussian

$$p(x[n]; \mu) = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}(x[n] - \mu)^2\right].$$

Find the BLUE of the mean μ in both cases. What can you say about the MVU estimator for μ?

6.5 The observed samples $\{x[0], x[1], \ldots, x[N-1]\}$ are IID according to the lognormal PDF

$$p(x[n]; \theta) = \begin{cases} \frac{1}{\sqrt{2\pi}x[n]} \exp\left[-\frac{1}{2}(\ln x[n] - \theta)^2\right] & x[n] > 0 \\ 0 & x[n] < 0. \end{cases}$$

Prove that the mean is $\exp(\theta + 1/2)$ and therefore that the unbiased constraint cannot be satisfied. Using the transformation of random variables approach with $y[n] = \ln x[n]$, find the BLUE for θ.

6.6 In this problem we extend the scalar BLUE results. Assume that $E(x[n]) = \theta s[n] + \beta$, where θ is the unknown parameter to be estimated and β is a *known* constant. The data vector $\mathbf{x}$ has covariance matrix $\mathbf{C}$. We define a modified linear (actually affine) estimator for this problem as

$$\hat{\theta} = \sum_{n=0}^{N-1} a_n x[n] + b.$$

Prove that the BLUE is given by

$$\hat{\theta} = \frac{\mathbf{s}^T \mathbf{C}^{-1}(\mathbf{x} - \beta \mathbf{1})}{\mathbf{s}^T \mathbf{C}^{-1} \mathbf{s}}.$$

Also, find the minimum variance.

6.7 Assume that $x[n] = As[n] + w[n]$ for $n = 0, 1, \ldots, N-1$ are observed, where $w[n]$ is zero mean noise with covariance matrix $\mathbf{C}$ and $s[n]$ is a known signal. The amplitude A is to be estimated using a BLUE. Find the BLUE and discuss what happens if $\mathbf{s} = [s[0]\, s[1] \ldots s[N-1]]^T$ is an eigenvector of $\mathbf{C}$. Also, find the minimum variance.

6.8 Continuing Problem 6.7, we can always represent the signal vector $\mathbf{s}$ as a linear combination of the eigenvectors of $\mathbf{C}$. This is because we can always find N eigenvectors that are linearly independent. Furthermore, it can be shown that

these eigenvectors are orthonormal due to the symmetric nature of $\mathbf{C}$. Hence, an orthogonal representation of the signal is

$$\mathbf{s} = \sum_{i=0}^{N-1} \alpha_i \mathbf{v}_i$$

where $\{\mathbf{v}_0, \mathbf{v}_1, \ldots, \mathbf{v}_{N-1}\}$ are the orthonormal eigenvectors of $\mathbf{C}$. Prove that the minimum variance of the BLUE is

$$\text{var}(\hat{A}) = \frac{1}{\sum_{i=0}^{N-1} \frac{\alpha_i^2}{\lambda_i}}$$

where λ_i is the eigenvalue of $\mathbf{C}$ corresponding to the eigenvector $\mathbf{v}_i$. Next show that the signal energy $\mathcal{E} = \mathbf{s}^T\mathbf{s}$ is given by $\sum_{i=0}^{N-1} \alpha_i^2$. Finally, prove that if the energy is constrained to be some value $\mathcal{E}_0$, then the smallest possible variance of the BLUE is obtained by choosing the signal

$$\mathbf{s} = c\mathbf{v}_{\min}$$

where $\mathbf{v}_{\min}$ is the eigenvector of $\mathbf{C}$ corresponding to the minimum eigenvalue and c is chosen to satisfy the energy constraint $\mathcal{E}_0 = \mathbf{s}^T\mathbf{s} = c^2$. Assume that the eigenvalues of $\mathbf{C}$ are distinct. Explain why this result makes sense.

6.9 In an on-off keyed (OOK) communication system we transmit one of two signals

$$s_0(t) = 0 \qquad 0 \leq t \leq T$$

to represent a binary 0 or

$$s_1(t) = A\cos 2\pi f_1 t \qquad 0 \leq t \leq T$$

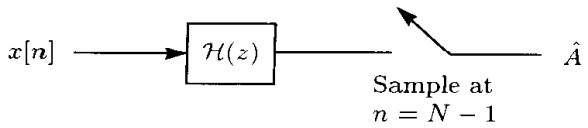
to represent a binary 1. The amplitude A is assumed to be positive. To determine which bit was transmitted we sample the received waveform to produce

$$x[n] = s_i[n] + w[n] \qquad n = 0, 1, \ldots, N-1.$$

If a 0 was transmitted, we would have $s_i[n] = 0$, and if a 1 was transmitted, we would have $s_i[n] = A\cos 2\pi f_1 n$ (assuming a sampling rate of 1 sample/s). The noise samples given by $w[n]$ are used to model channel noise. A simple receiver might estimate the sinusoidal amplitude A and decide a **1** was sent if $\hat{A} > \gamma$, and a 0 if $\hat{A} < \gamma$. Hence, an equivalent problem is to estimate the amplitude A for the received data

$$x[n] = A\cos 2\pi f_1 n + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is zero mean noise with covariance matrix $\mathbf{C} = \sigma^2\mathbf{I}$. Find the BLUE for this problem and interpret the resultant detector. Find the best frequency in the range $0 \leq f_1 < 1/2$ to use at the transmitter.



$$\mathcal{H}(z) = \sum_{k=0}^{N-1} h[k]z^{-k}$$

Figure 6.5 FIR filter estimator for Problem 6.11

6.10 We continue Problem 6.9 by examining the question of signal selection for an OOK system in the presence of *colored* noise. Let the noise $w[n]$ be a zero mean WSS random process with ACF

$$r_{ww}[k] = \begin{cases} 1.81 & k = 0 \\ 0 & k = 1 \\ 0.9 & k = 2 \\ 0 & k \geq 3. \end{cases}$$

Find the PSD and plot it for frequencies $0 \leq f \leq 1/2$. As in the previous problem, find the frequency which yields the smallest value of the BLUE variance for $N = 50$. Explain your results. Hint: You will need a computer to do this.

6.11 Consider the problem of estimating a DC level in WSS noise or given

$$x[n] = A + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is a zero mean WSS random process with ACF $r_{ww}[k]$, estimate A. It is proposed to estimate A by using the output of the FIR filter shown in Figure 6.5 at time $n = N - 1$. Note that the estimator is given by

$$\hat{A} = \sum_{k=0}^{N-1} h[k]x[N-1-k].$$

The input $x[n]$ is assumed to be zero for $n < 0$. To obtain a good estimator we wish to have the filter pass the DC signal A but block the noise $w[n]$. Hence, we choose $H(\exp(j0)) = 1$ as a constraint and minimize the noise power at the output at time $n = N - 1$ by our choice of the FIR filter coefficients $h[k]$. Find the optimal filter coefficients and the minimum noise power at the output of the optimal filter. Explain your results.

6.12 Prove that the BLUE commutes over linear (actually affine) transformations of $\boldsymbol{\theta}$. Thus, if we wish to estimate

$$\boldsymbol{\alpha} = \mathbf{B}\boldsymbol{\theta} + \mathbf{b}$$

where $\mathbf{B}$ is a known $p \times p$ invertible matrix and $\mathbf{b}$ is a known $p \times 1$ vector, prove that the BLUE is given by

$$\hat{\boldsymbol{\alpha}} = \mathbf{B}\hat{\boldsymbol{\theta}} + \mathbf{b}$$

where $\hat{\boldsymbol{\theta}}$ is the BLUE for $\boldsymbol{\theta}$. Assume that $\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w}$, where $E(\mathbf{w}) = \mathbf{0}$ and $E(\mathbf{w}\mathbf{w}^T) = \mathbf{C}$. Hint: Replace $\boldsymbol{\theta}$ in the data model by $\boldsymbol{\alpha}$.

6.13 In this problem it is shown that the BLUE is identical to the *weighted least squares estimator* to be discussed in Chapter 8. In particular, the weighted least squares estimator is found by minimizing the criterion

$$J = (\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T \mathbf{C}^{-1} (\mathbf{x} - \mathbf{H}\boldsymbol{\theta}).$$

Prove that the $\boldsymbol{\theta}$ that minimizes J is the BLUE.

6.14 A noise process is composed of IID zero mean random variables with PDF

$$p(w[n]) = \frac{1-\epsilon}{\sqrt{2\pi\sigma_B^2}} \exp\left[-\frac{1}{2}\left(\frac{w^2[n]}{\sigma_b^2}\right)\right] + \frac{\epsilon}{\sqrt{2\pi\sigma_I^2}} \exp\left[-\frac{1}{2}\left(\frac{w^2[n]}{\sigma_I^2}\right)\right]$$

where $0 < \epsilon < 1$. Such a PDF is called a Gaussian mixture. It is used to model noise that is Gaussian with variance σ_B^2 for $100(1-\epsilon)\%$ of the time and Gaussian with variance σ_I^2 the remaining time. Typically, $\sigma_I^2 \gg \sigma_B^2$ and $\epsilon \ll 1$, so that the noise is predominately due to *background* noise with variance σ_B^2 but also contains occasional high level events or *interferences* modeled as Gaussian noise with variance σ_I^2. Show that the variance of this PDF is

$$\sigma^2 = (1-\epsilon)\sigma_B^2 + \epsilon\sigma_I^2.$$

If $\{w^2[0], w^2[1], \ldots, w^2[N-1]\}$ is considered to be the data and σ_B^2, ϵ are assumed known, find the BLUE of σ_I^2. Hint: Use the results of Problem 6.12.

6.15 For the general linear model

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{s} + \mathbf{w}$$

where $\mathbf{s}$ is a known $N \times 1$ vector and $E(\mathbf{w}) = \mathbf{0}$, $E(\mathbf{w}\mathbf{w}^T) = \mathbf{C}$, find the BLUE.

6.16 In implementing the BLUE for the linear model described in Theorem 6.1, we may unknowingly use the wrong covariance matrix to form the estimate

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T \hat{\mathbf{C}}^{-1} \mathbf{H})^{-1} \mathbf{H}^T \hat{\mathbf{C}}^{-1} \mathbf{x}.$$

To examine the effect of this modeling error assume that we wish to estimate A as in Example 6.2. Find the variance of this estimator and that of the BLUE for the case where $N = 2$ and

$$\mathbf{C} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\hat{\mathbf{C}} = \begin{bmatrix} 1 & 0 \\ 0 & \alpha \end{bmatrix}.$$

Compare the variances for $\alpha \to 0$, $\alpha = 1$, and $\alpha \to \infty$.

Appendix 6A

Derivation of Scalar BLUE

To minimize $\text{var}(\hat{\theta}) = \mathbf{a}^T\mathbf{C}\mathbf{a}$ subject to the constraint $\mathbf{a}^T\mathbf{s} = 1$ we use the method of Lagrangian multipliers. The Lagrangian function J becomes

$$J = \mathbf{a}^T\mathbf{C}\mathbf{a} + \lambda(\mathbf{a}^T\mathbf{s} - 1).$$

Using (4.3), the gradient with respect to $\mathbf{a}$ is

$$\frac{\partial J}{\partial \mathbf{a}} = 2\mathbf{C}\mathbf{a} + \lambda\mathbf{s}.$$

Setting this equal to the zero vector and solving produces

$$\mathbf{a} = -\frac{\lambda}{2}\mathbf{C}^{-1}\mathbf{s}.$$

The Lagrangian multiplier λ is found using the constraint equation as

$$\mathbf{a}^T\mathbf{s} = -\frac{\lambda}{2}\mathbf{s}^T\mathbf{C}^{-1}\mathbf{s} = 1$$

or

$$-\frac{\lambda}{2} = \frac{1}{\mathbf{s}^T\mathbf{C}^{-1}\mathbf{s}}$$

so that the gradient is zero with the constraint satisfied for

$$\mathbf{a}_{\text{opt}} = \frac{\mathbf{C}^{-1}\mathbf{s}}{\mathbf{s}^T\mathbf{C}^{-1}\mathbf{s}}.$$

The variance for this value of $\mathbf{a}$ is found from

$$\begin{aligned} \text{var}(\hat{\theta}) &= \mathbf{a}_{\text{opt}}^T\mathbf{C}\mathbf{a}_{\text{opt}} \\ &= \frac{\mathbf{s}^T\mathbf{C}^{-1}\mathbf{C}\mathbf{C}^{-1}\mathbf{s}}{(\mathbf{s}^T\mathbf{C}^{-1}\mathbf{s})^2} \\ &= \frac{1}{\mathbf{s}^T\mathbf{C}^{-1}\mathbf{s}}. \end{aligned}$$

That $\mathbf{a}_{\text{opt}}$ is indeed the global minimum and not just a local minimum is verified by considering the function

$$G = (\mathbf{a} - \mathbf{a}_{\text{opt}})^T \mathbf{C} (\mathbf{a} - \mathbf{a}_{\text{opt}}).$$

Expanding G, we have

$$\begin{aligned} G &= \mathbf{a}^T \mathbf{C} \mathbf{a} - 2\mathbf{a}_{\text{opt}}^T \mathbf{C} \mathbf{a} + \mathbf{a}_{\text{opt}}^T \mathbf{C} \mathbf{a}_{\text{opt}} \\ &= \mathbf{a}^T \mathbf{C} \mathbf{a} - 2\frac{\mathbf{s}^T \mathbf{C}^{-1} \mathbf{C} \mathbf{a}}{\mathbf{s}^T \mathbf{C}^{-1} \mathbf{s}} + \frac{1}{\mathbf{s}^T \mathbf{C}^{-1} \mathbf{s}} \\ &= \mathbf{a}^T \mathbf{C} \mathbf{a} - \frac{1}{\mathbf{s}^T \mathbf{C}^{-1} \mathbf{s}} \end{aligned}$$

where we have used the constraint equation. Hence,

$$\mathbf{a}^T \mathbf{C} \mathbf{a} = (\mathbf{a} - \mathbf{a}_{\text{opt}})^T \mathbf{C} (\mathbf{a} - \mathbf{a}_{\text{opt}}) + \frac{1}{\mathbf{s}^T \mathbf{C}^{-1} \mathbf{s}}$$

which is uniquely minimized for $\mathbf{a} = \mathbf{a}_{\text{opt}}$. This is because the positive definite property of $\mathbf{C}$ makes the first term on the right-hand side greater than zero for all $(\mathbf{a} - \mathbf{a}_{\text{opt}}) \neq \mathbf{0}$.

Appendix 6B

Derivation of Vector BLUE

To derive the BLUE for a vector parameter we need to minimize

$$\text{var}(\hat{\theta}_i) = \mathbf{a}_i^T \mathbf{C} \mathbf{a}_i$$

for $i = 1, 2, \ldots, p$ subject to the constraints

$$\mathbf{a}_i^T \mathbf{h}_j = \delta_{ij} \qquad i = 1, 2, \ldots, p; j = 1, 2, \ldots, p.$$

This problem is much the same as that of the scalar BLUE (see Appendix 6A) except for the additional constraints on $\mathbf{a}_i$. Now we have p constraints on $\mathbf{a}_i$ instead of just one. Since each $\mathbf{a}_i$ is free to assume any value, independently of the others, we actually have p separate minimization problems linked only by the constraints. Proceeding in a similar manner to that of Appendix 6A, we consider the Lagrangian function for $\mathbf{a}_i$.

$$J_i = \mathbf{a}_i^T \mathbf{C} \mathbf{a}_i + \sum_{j=1}^{p} \lambda_j^{(i)} (\mathbf{a}_i^T \mathbf{h}_j - \delta_{ij}).$$

Taking the gradient of the Lagrangian using (4.3)

$$\frac{\partial J_i}{\partial \mathbf{a}_i} = 2\mathbf{C}\mathbf{a}_i + \sum_{j=1}^{p} \lambda_j^{(i)} \mathbf{h}_j.$$

We now let $\boldsymbol{\lambda}_i = [\lambda_1^{(i)} \lambda_2^{(i)} \ldots \lambda_p^{(i)}]^T$ and note that $\mathbf{H} = [\mathbf{h}_1 \mathbf{h}_2 \ldots \mathbf{h}_p]$ to yield

$$\frac{\partial J_i}{\partial \mathbf{a}_i} = 2\mathbf{C}\mathbf{a}_i + \mathbf{H}\boldsymbol{\lambda}_i.$$

Setting the gradient equal to zero yields

$$\mathbf{a}_i = -\frac{1}{2}\mathbf{C}^{-1}\mathbf{H}\boldsymbol{\lambda}_i. \tag{6B.1}$$

To find the vector of Lagrangian multipliers we use the constraint equations

$$\mathbf{a}_i^T \mathbf{h}_j = \delta_{ij} \qquad j = 1, 2, \ldots, p$$

or in combined form

$$\begin{bmatrix} \mathbf{h}_1^T \\ \vdots \\ \mathbf{h}_{i-1}^T \\ \mathbf{h}_i^T \\ \mathbf{h}_{i+1}^T \\ \vdots \\ \mathbf{h}_p^T \end{bmatrix} \mathbf{a}_i = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Letting $\mathbf{e}_i$ denote the vector of all zeros except in the ith place, we have the constraint equation

$$\mathbf{H}^T \mathbf{a}_i = \mathbf{e}_i.$$

Using (6B.1), the constraint equations become

$$\mathbf{H}^T \mathbf{a}_i = -\frac{1}{2} \mathbf{H}^T \mathbf{C}^{-1} \mathbf{H} \boldsymbol{\lambda}_i = \mathbf{e}_i$$

so that assuming invertibility of $\mathbf{H}^T \mathbf{C}^{-1} \mathbf{H}$, the Lagrangian multiplier vector is

$$-\frac{1}{2} \boldsymbol{\lambda}_i = (\mathbf{H}^T \mathbf{C}^{-1} \mathbf{H})^{-1} \mathbf{e}_i.$$

Using this in (6B.1) yields the final result

$$\mathbf{a}_{i_{\text{opt}}} = \mathbf{C}^{-1} \mathbf{H} (\mathbf{H}^T \mathbf{C}^{-1} \mathbf{H})^{-1} \mathbf{e}_i \tag{6B.2}$$

and a corresponding variance of

$$\begin{aligned} \text{var}(\hat{\theta}_i) &= \mathbf{a}_{i_{\text{opt}}}^T \mathbf{C} \mathbf{a}_{i_{\text{opt}}} \\ &= \mathbf{e}_i^T (\mathbf{H}^T \mathbf{C}^{-1} \mathbf{H})^{-1} \mathbf{H}^T \mathbf{C}^{-1} \mathbf{C} \mathbf{C}^{-1} \mathbf{H} (\mathbf{H}^T \mathbf{C}^{-1} \mathbf{H})^{-1} \mathbf{e}_i \\ &= \mathbf{e}_i^T (\mathbf{H}^T \mathbf{C}^{-1} \mathbf{H})^{-1} \mathbf{e}_i. \end{aligned}$$

In like fashion to Appendix 6A, $\mathbf{a}_{i_{\text{opt}}}$ can be shown to produce the global minimum. We can express the vector BLUE in a more compact form by letting

$$\begin{aligned} \hat{\boldsymbol{\theta}} &= \begin{bmatrix} \mathbf{a}_{1_{\text{opt}}}^T \mathbf{x} \\ \mathbf{a}_{2_{\text{opt}}}^T \mathbf{x} \\ \vdots \\ \mathbf{a}_{p_{\text{opt}}}^T \mathbf{x} \end{bmatrix} = \begin{bmatrix} \mathbf{e}_1^T (\mathbf{H}^T \mathbf{C}^{-1} \mathbf{H})^{-1} \mathbf{H}^T \mathbf{C}^{-1} \mathbf{x} \\ \mathbf{e}_2^T (\mathbf{H}^T \mathbf{C}^{-1} \mathbf{H})^{-1} \mathbf{H}^T \mathbf{C}^{-1} \mathbf{x} \\ \vdots \\ \mathbf{e}_p^T (\mathbf{H}^T \mathbf{C}^{-1} \mathbf{H})^{-1} \mathbf{H}^T \mathbf{C}^{-1} \mathbf{x} \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{e}_1^T \\ \mathbf{e}_2^T \\ \vdots \\ \mathbf{e}_p^T \end{bmatrix} (\mathbf{H}^T \mathbf{C}^{-1} \mathbf{H})^{-1} \mathbf{H}^T \mathbf{C}^{-1} \mathbf{x} \\ &= (\mathbf{H}^T \mathbf{C}^{-1} \mathbf{H})^{-1} \mathbf{H}^T \mathbf{C}^{-1} \mathbf{x} \end{aligned}$$

since $[\mathbf{e}_1\, \mathbf{e}_2 \ldots \mathbf{e}_p]^T$ is the identity matrix. Also, the covariance matrix of $\hat{\boldsymbol{\theta}}$ is

$$\mathbf{C}_{\hat{\theta}} = E\left[(\hat{\boldsymbol{\theta}} - E(\hat{\boldsymbol{\theta}}))(\hat{\boldsymbol{\theta}} - E(\hat{\boldsymbol{\theta}}))^T\right]$$

where

$$\begin{aligned} \hat{\boldsymbol{\theta}} - E(\hat{\boldsymbol{\theta}}) &= (\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{C}^{-1}(\mathbf{H}\boldsymbol{\theta} + \mathbf{w}) - E(\hat{\boldsymbol{\theta}}) \\ &= (\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{C}^{-1}\mathbf{w}. \end{aligned}$$

Thus,

$$\mathbf{C}_{\hat{\theta}} = E[(\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{C}^{-1}\mathbf{w}\mathbf{w}^T\mathbf{C}^{-1}\mathbf{H}(\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}].$$

Since the covariance matrix of $\mathbf{w}$ is $\mathbf{C}$, we have

$$\begin{aligned} \mathbf{C}_{\hat{\theta}} &= (\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{C}^{-1}\mathbf{C}\mathbf{C}^{-1}\mathbf{H}(\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1} \\ &= (\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1} \end{aligned}$$

with the minimum variances given by the diagonal elements of $\mathbf{C}_{\hat{\theta}}$ or

$$\mathrm{var}(\hat{\theta}_i) = \mathbf{e}_i^T\mathbf{C}_{\hat{\theta}}\mathbf{e}_i = [(\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}]_{ii}$$

as previously derived.